

SQL Case Study — Part 2

Restaurant Analytics — Jomato Dataset

Continuing the restaurant analytics case study — this round covers stored procedures, transactions, window functions, loops, views, and triggers.

- Stored Procedures for Reusable Reporting Logic
- Transactions & ROLLBACK for Safe Data Updates
- ROW_NUMBER() Window Function for Ranked Results
- WHILE Loops for Procedural Control Flow
- Views to Store Reusable Query Results
- Triggers for Automated Post-Insert Actions

Step 1: Problem Statement & Tasks

Dataset: Jomato

About the dataset:

You work for a data analytics company, and your client is a food delivery platform similar to Jomato. They have provided you with a dataset containing information about various restaurants in a city. Your task is to analyze this dataset using SQL queries to extract valuable insights and generate reports for your client.

Tasks to be performed:

1. Create a stored procedure to display the restaurant name, type and cuisine where the table booking is not zero.
2. Create a transaction and update the cuisine type 'Cafe' to 'Cafeteria'. Check the result and rollback it.
3. Generate a row number column and find the top 5 areas with the highest rating of restaurants.
4. Use the while loop to display the 1 to 50.
5. Write a query to Create a Top rating view to store the generated top 5 highest rating of restaurants.
6. Create a trigger that give an message whenever a new record is inserted.

Step 2: Queries 1 & 2: Stored Procedure (Table Booking Filter) + Transaction with ROLLBACK

```
--1 Create a stored procedure to display the restaurant name, type and  
--cuisine where the table booking is not zero.
```

```
USE JomatoDB;
```

```
GO
```

```
CREATE PROCEDURE GetTableBookingRestaurants
```

```
AS
```

```
BEGIN
```

```
    SELECT
```

```
        RestaurantName, |
```

```
        RestaurantType,
```

```
        CuisinesType
```

```
    FROM dbo.Jomato
```

```
    WHERE TableBooking <> 0;
```

```
END;
```

```
GO
```

```
EXEC GetTableBookingRestaurants;
```

```
--2. Create a transaction and update the cuisine type 'Cafe'
```

```
--to 'Cafeteria'. Check the result and rollback it.;
```

```
USE JomatoDB;
```

```
GO
```

```
BEGIN TRANSACTION;
```

```
UPDATE dbo.Jomato
```

```
SET CuisinesType = 'Cafeteria'
```

```
WHERE CuisinesType = 'Cafe';
```

```
SELECT *
```

```
FROM dbo.Jomato
```

```
WHERE CuisinesType = 'Cafeteria';
```

```
ROLLBACK TRANSACTION;
```

Step 3: Queries 3 & 4: ROW_NUMBER() for Top-Rated Areas + WHILE Loop (1 to 50)

```
--3.Generate a row number column and find the top 5 areas with
--the highest rating of restaurants.
USE JomatoDB;
GO

SELECT TOP 5
    ROW_NUMBER() OVER (ORDER BY MAX(Rating) DESC) AS RowNumber,
    Area,
    MAX(Rating) AS HighestRating
FROM dbo.Jomato
GROUP BY Area
ORDER BY HighestRating DESC;

--4.Use the while loop to display the 1 to 50.
DECLARE @Number INT = 1;

WHILE @Number <= 50
BEGIN
    PRINT @Number;
    SET @Number = @Number + 1;
END;
```

Step 4: Queries 5 & 6: View for Top 5 Ratings + Trigger on New Insert

--5. Write a query to Create a Top rating view to store the
--generated top 5 highest rating of restaurants.

```
USE JomatoDB;  
GO
```

```
CREATE VIEW TopRating  
AS  
SELECT TOP 5  
    RestaurantName,  
    RestaurantType,  
    CuisinesType,  
    Rating  
FROM dbo.Jomato  
ORDER BY Rating DESC;  
GO
```

```
select * from TopRating;
```

--6. Create a trigger that give an message whenever a new record is inserted.

```
|  
CREATE TRIGGER NewRecordMessage  
ON dbo.Jomato  
AFTER INSERT  
AS  
BEGIN  
    PRINT 'New record has been inserted successfully.';  
END;  
GO
```